Exam 2 on Friday!

- Bring a calculator!
- around 10-12 questions, some with a few parts
  - a couple short answer questions at start
  - then the rest are step-by-step data analysis for a given data set (very similar format to Exam 1)

Equations to know:

- R^2 and R^2_{adj}
- VIF $\leftrightarrow$ R^2
- t statistic $\leftarrow$ $t = \dfrac{\hat{\beta}_j - (\text{value under } H_0)}{SE_{\hat{\beta}_j}}$
- F statistic $\leftarrow$ $\dfrac{(SSE_{reduced} - SSE_{full}) / (p_{full} - p_{reduced})}{\frac{1}{n-p_{full}} SSE_{full}}$
- Fill in ANOVA table
- SSE, SSTotal, SSModel

$$SSE = \sum_{i=1}^{n}(Y_i - \hat{Y}_i)^2, \quad SSTotal = \sum_{i=1}^{n}(Y_i - \bar{Y})^2, \quad SSModel \rightarrow \sum_{i=1}^{n}(\hat{Y}_i - \bar{Y})^2$$

Q 18 on review

$$F = \frac{(SSE_{reduced} - SSE_{full}) / (P_{full} - P_{reduced})}{SSE_{full} / (n - P_{full})}$$

$n = 100$

full model: $\beta_0 + \beta_1 \text{mass} + \beta_2 \text{lsShip} + \beta_3 \text{mass.lsShip} + \varepsilon$

$P_{full} = 4 \Rightarrow n - P_{full} = 96$

reduced model: $H_0: \beta_2, \beta_3 = 0$ H_A: at least one of $\beta_2, \beta_3 \neq 0$

$\Rightarrow \beta_0 + \beta_1 \text{mass} + \varepsilon$

$P_{reduced} = 2 \Rightarrow P_{full} - P_{reduced} = 2$

SSE_{full} : "residuals" "Sum Sq" of anova output

26192

$SSE_{reduced}$

163817

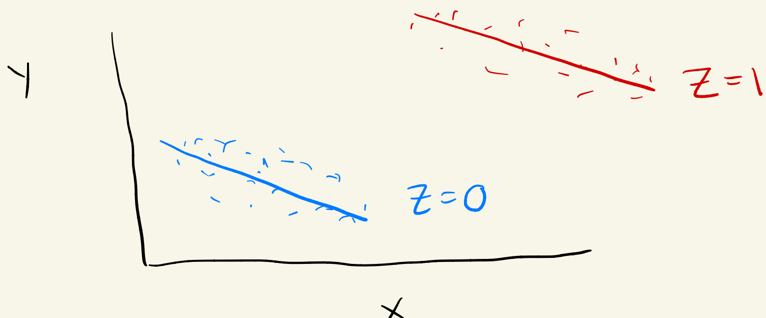
$$F = \frac{(163817 - 26192)/2}{26192/96} = 252.2$$

Simpson's paradox

observed relationship changes when we look inside groups

just looking at X & Y:
appears to be a positive relationship between X & Y

$Y=\beta_0+\beta_1 X+\varepsilon$



once we account for group variable Z, negative relationship between X & Y

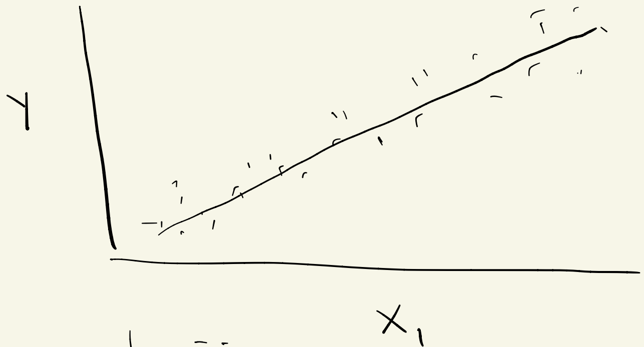
(This happens because Z is also related to both X & Y
$Z=1 \rightarrow$ higher X, higher Y)

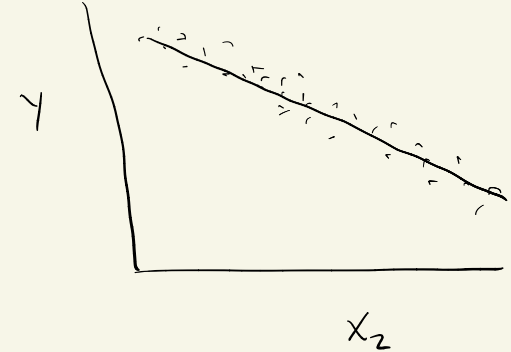
$Y=\beta_0+\beta_1 X+\beta_2 Z+\varepsilon$

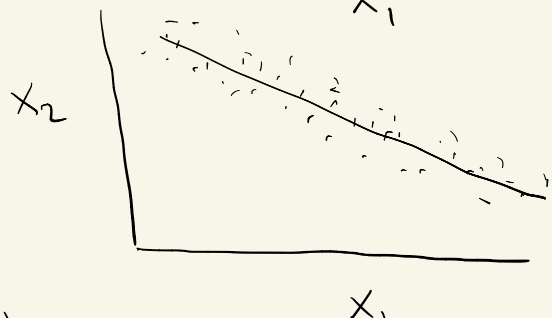
$Y=\beta_0+\beta_1 X+\beta_2 Z+\beta_3 X\cdot Z+\varepsilon$

VIFs

X_1, X_2, Y want to predict Y







Causes problems for fitting model

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \varepsilon$$

interpretation: β_2: a one-unit increase in X_2 is associated w/ an average change of β_2 in Y, holding X_1 fixed

problem when X_1, X_2 are closely related b/c can't fix one & change the other

Multicollinearity also increases variability of $\hat{\beta}_1$ & $\hat{\beta}_2$

VIF: how much does variability increase?

$$\text{VIF}(X_1) = \frac{\text{Variance}(\hat{\beta}_1) \text{ when } X_1, X_2 \text{ in model}}{\text{Variance}(\hat{\beta}_1) \text{ when only } X_1 \text{ in model}}$$

$$= \frac{1}{1 - R^2(X_1)}$$

$R^2(X_1) = R^2$ for predicting X_1 from X_2

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \beta_4 X_4 + \beta_5 X_5 + \varepsilon$$

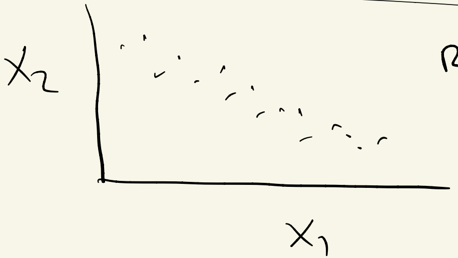
5 different VIFs:

$$VIF_1 = \frac{1}{1-R^2(X_1)} \leftarrow$$ R^2 for predicting X_1 from $X_2, \ldots, X_5$

e.g. $X_1 = \alpha_0 + \alpha_1 X_2 + \alpha_2 X_3 \cdots + \alpha_4 X_5 + \varepsilon$

$$VIF_2 = \frac{1}{1-R^2(X_2)} \leftarrow$$ R^2 for predicting X_2 from X_1, X_3, X_4, X_5

$$\Rightarrow 1 - R^2(X_2) = \frac{1}{VIF_2} = 1 - \frac{1}{VIF_2} = R^2(X_2)$$

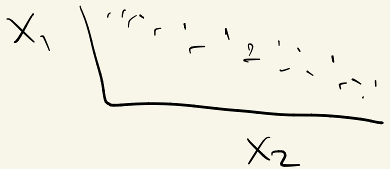


$R^2 = r^2$

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \varepsilon$$

$$VIF_1 = \frac{1}{1-R^2(X_1)}$$

$$VIF_2 = \frac{1}{1-R^2(X_2)}$$



F & t distributions

t test:
$$t = \frac{\hat{\beta}_j - (\text{value under null})}{SE_{\hat{\beta}_j}}$$

e.g. $H_0: \beta_j = 0$ $\quad t = \dfrac{\hat{\beta}_j}{SE_{\hat{\beta}_j}}$

numerator df

F statistic:
$$F = \frac{(SSE_{reduced} - SSE_{full}) / \overbrace{(p_{full} - p_{reduced})}^{\text{numerator df}}}{SSE_{full} / \underbrace{(n - p_{full})}_{\text{denominator df}}}$$

ratio of (sums of) squared things

denominator df

$$t^2 = \frac{\hat{\beta}_j^2}{(SE_{\beta_j})^2} = \frac{\hat{\beta}_j^2}{Var(\hat{\beta}_j)} = \text{ratio of squared things}$$

Turns out: $t^2 \sim F$ distribution

$$\boxed{t \sim t_{n-p} \quad \text{then} \quad t^2 \sim F_{1, n-p}}$$

F distribution is parameterized by two sets of degrees of freedom (numerator & denominator)

ANOVA Table (a single categorical explanatory variable)

$p = 3$ $\quad$ bill length $= \beta_0 + \beta_1 \text{ IsChinstrap} + \beta_2 \text{ IsGentoo} + \varepsilon$

want to test: $\quad H_0: \beta_1, \beta_2 = 0$ (no relationship between species & bill length)

$H_A:$ at least one of $\beta_1, \beta_2 \neq 0$ (some relationship between species & bill length, i.e. some difference in avg. bill length between species)

$$F = \frac{MSModel}{MSE} = \frac{SSModel/(p-1)}{SSE/(n-p)}$$

$n =$ # obs $\quad p =$ # parameters in model

Table

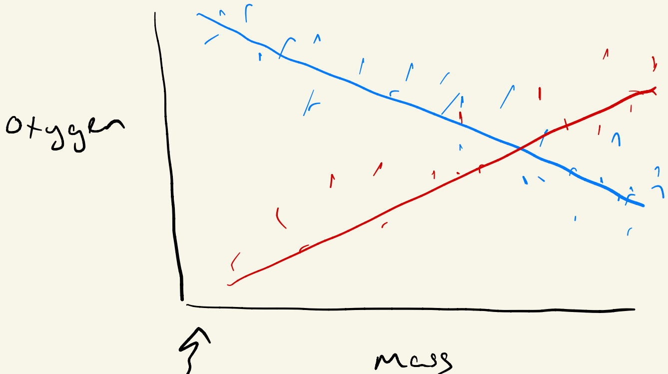
	df	SS	MS	F
Model	$p-1$	SSModel	MSModel	MSModel / MSE
Residuals	$n-p$	SSE	MSE	
Total	$n-1$	SSTotal		

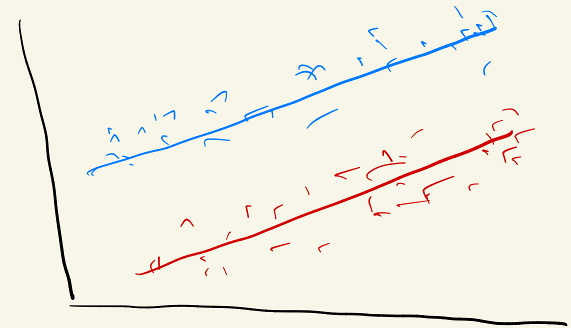
$SSTotal = SSModel + SSE \qquad MSModel = \dfrac{SSModel}{p-1} \qquad MSE = \dfrac{SSE}{n-p}$

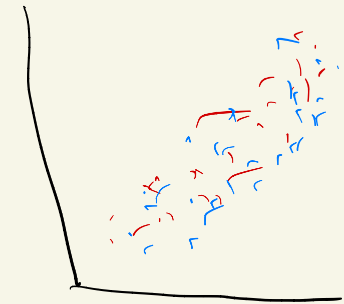
① Start by fill in df (from description of data, model (count # of parameters))

② I will give you two out of: SSModel, SSE, MSModel, MSE, F, SSTotal, Then fill in rest

Mass, Oxygen, Noise







(no difference in slope or intercept by Noise)

$$\text{Oxygen} = \beta_0 + \beta_1 \text{ mass} + \varepsilon$$

$$\text{oxygen} = \beta_0 + \beta_1 \text{ mass} + \beta_2 \text{ IsShip} + \varepsilon$$

$$\text{oxygen} = \beta_0 + \beta_1 \text{ mass} + \beta_2 \text{ IsShip} + \beta_3 \text{ mass} \cdot \text{IsShip} + \varepsilon$$